



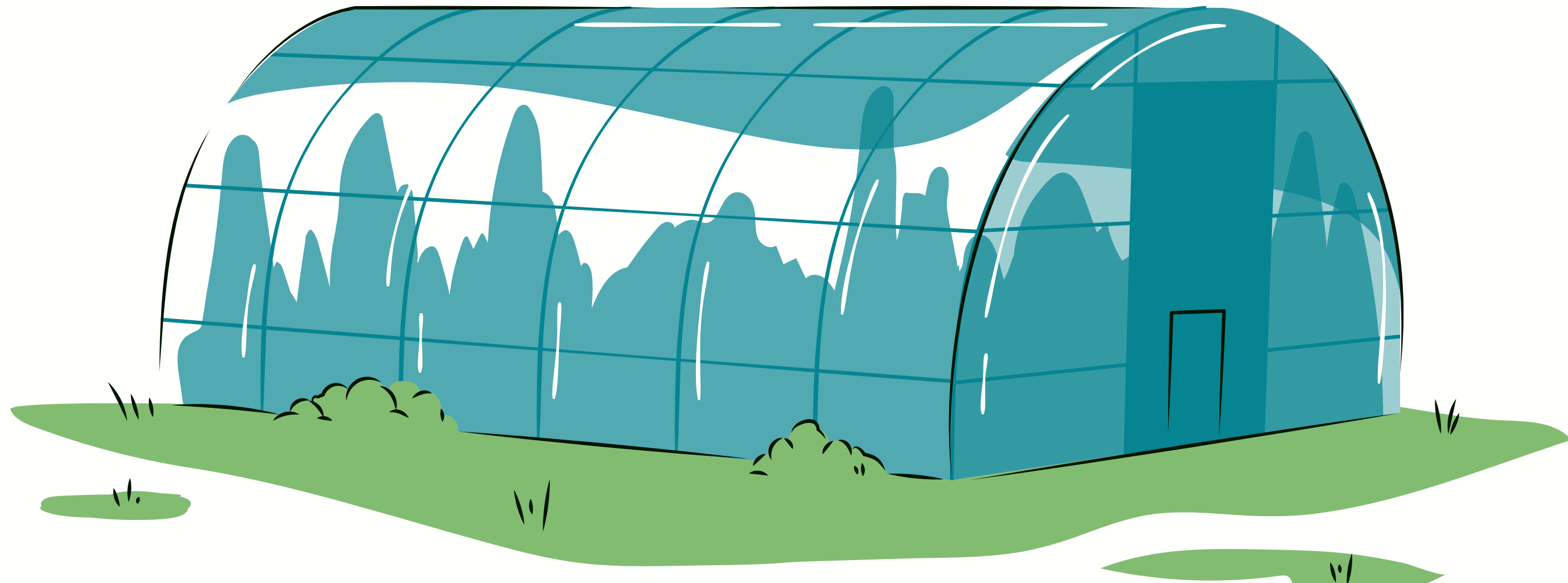
SMART GREEN HOUSE

Presented by

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Shahad Fahad Alshahwan 441500217

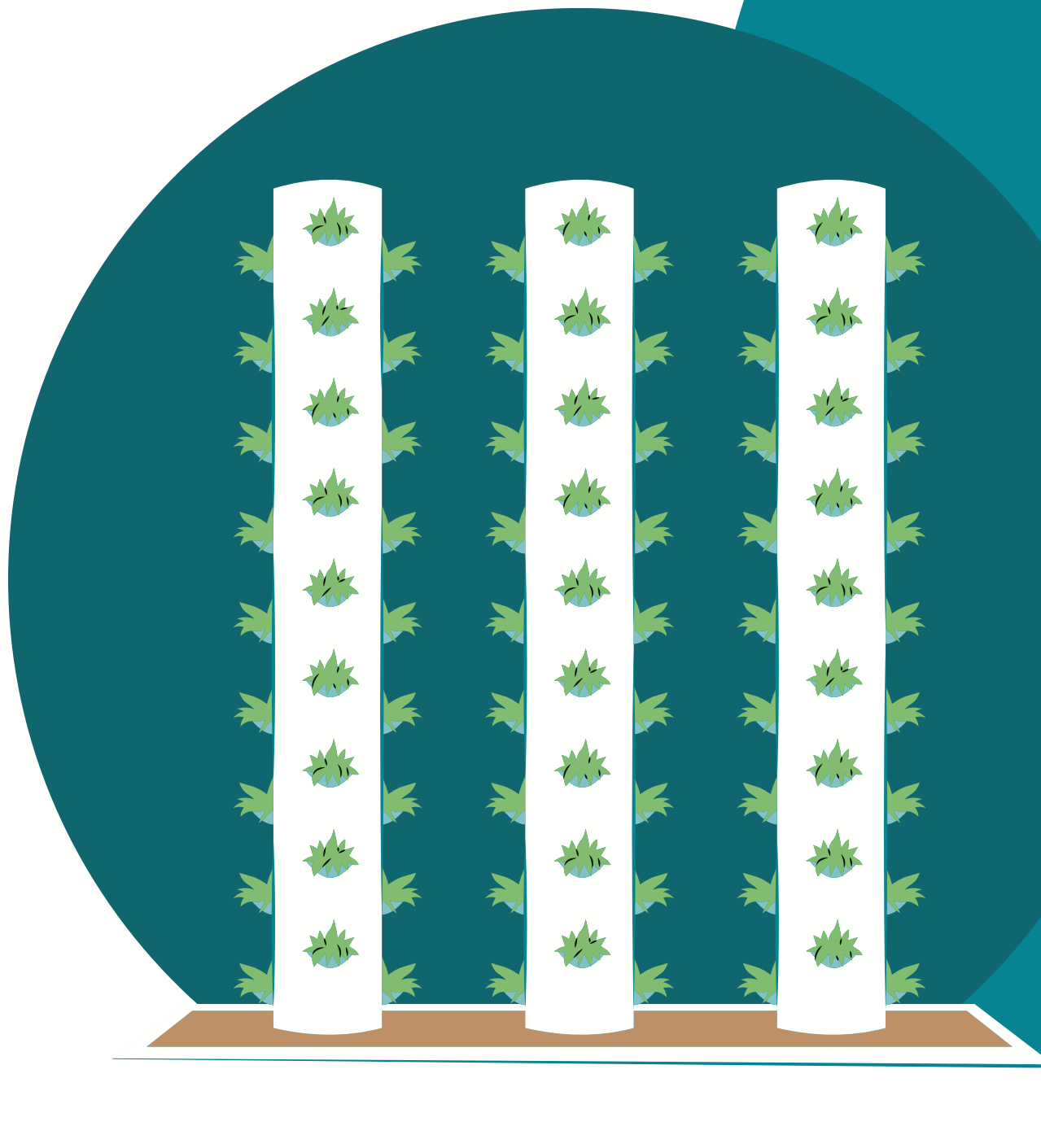
Amnah Ahmed 441500522



SMART GREENHOUSE SYSTEM

An automated IoT solution for optimal plant growth

- 🌡️ Monitors temperature, humidity & soil moisture
- 💧 Automatic irrigation using pump & servo motor
- 🔥 Fire detection with instant alerts
 - 📱 Remote monitoring via Blynk
 - ✅ Telegram notifications



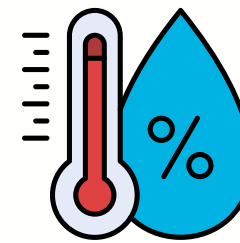
PROBLEM

Traditional greenhouses rely on manual checking and control.

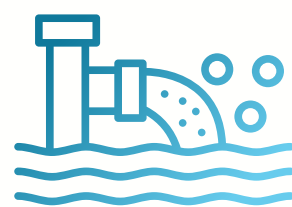
This causes:



Inefficient irrigation,
causing overwatering or
underwatering.



No real-time temperature and
humidity monitoring

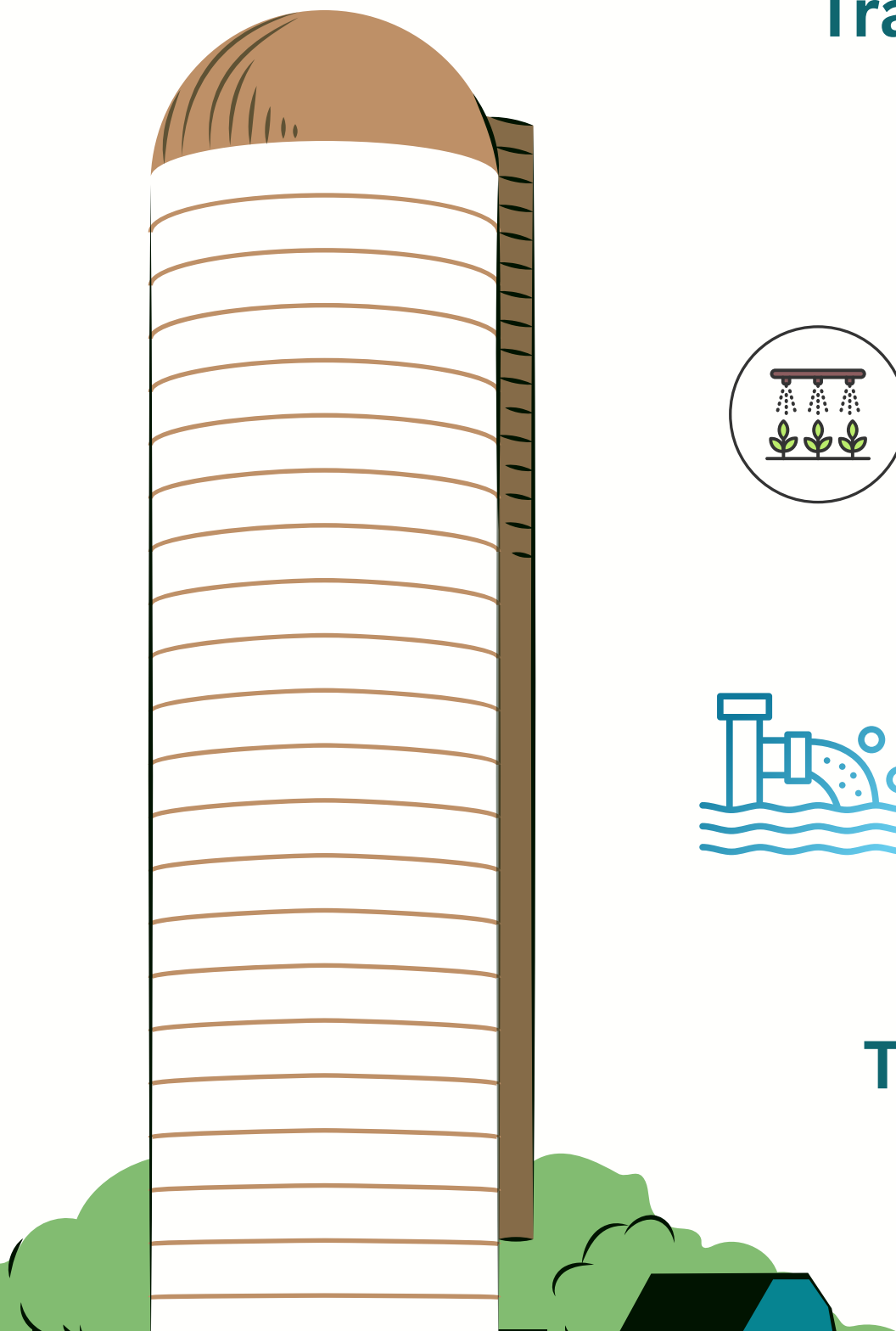


Water waste due to
manual irrigation



Delayed fire detection and
safety risks

**These issues negatively affect plant growth and reduce
resource efficiency.**



SOLUTION

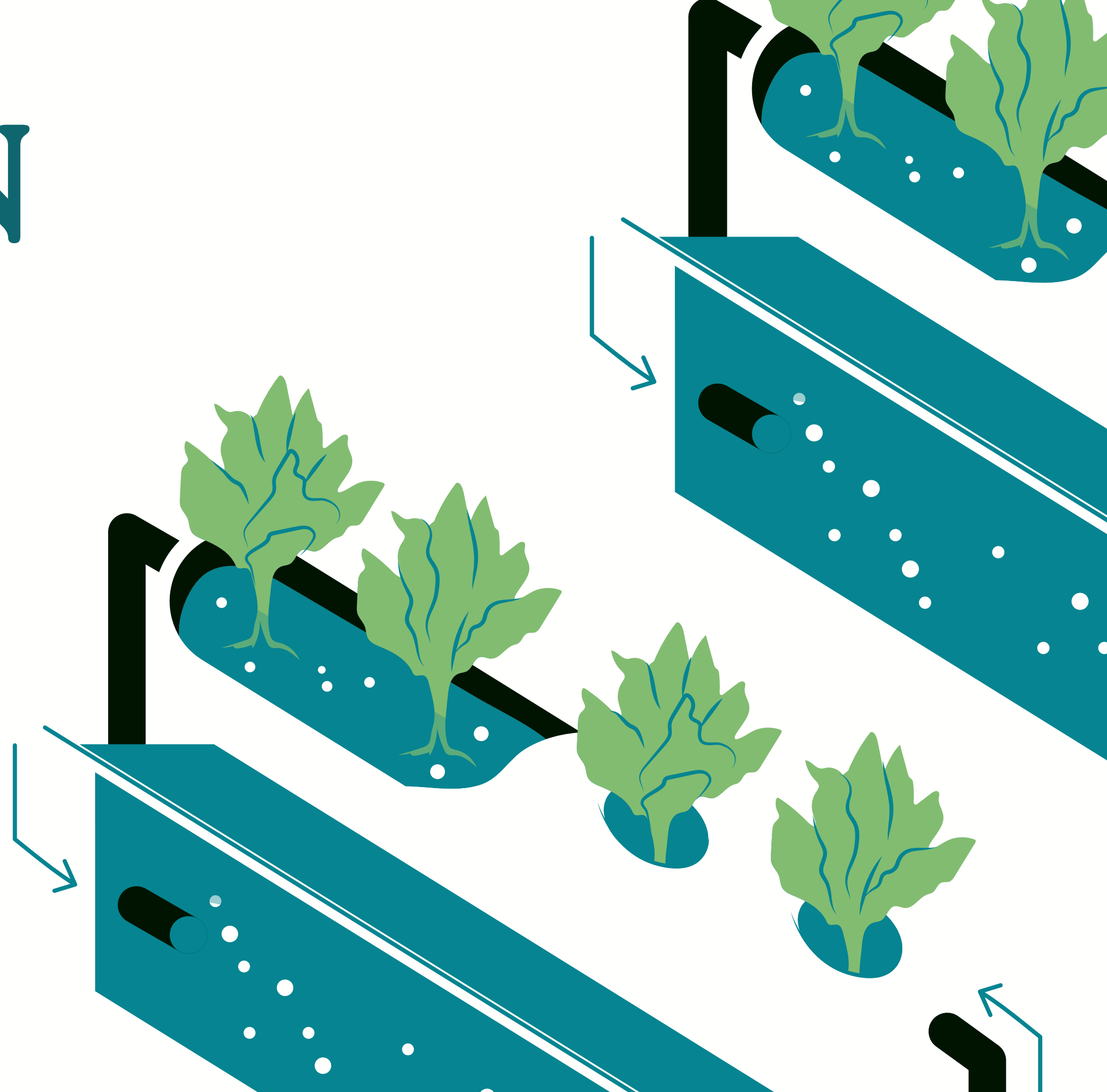


Check temperature, humidity, and soil wetness all the time.
If the soil is dry, **the water pump turns on by itself.**

If it gets too hot, **the fan turns on by itself.**



The system sends alerts if something is wrong.



TOOL USED

- ESP32 development board
- Blynk app
- micro usb
- Solderless Breadboard
- Male to Female & Male to Male jumper wires
- jumpers
- DHT11
- Humidity sensor
- 12v Pump
- DC 12v Fan
- Tube Hose pipe 6mm
- SG90 Rotation Servo Motor
- Flame Sensor

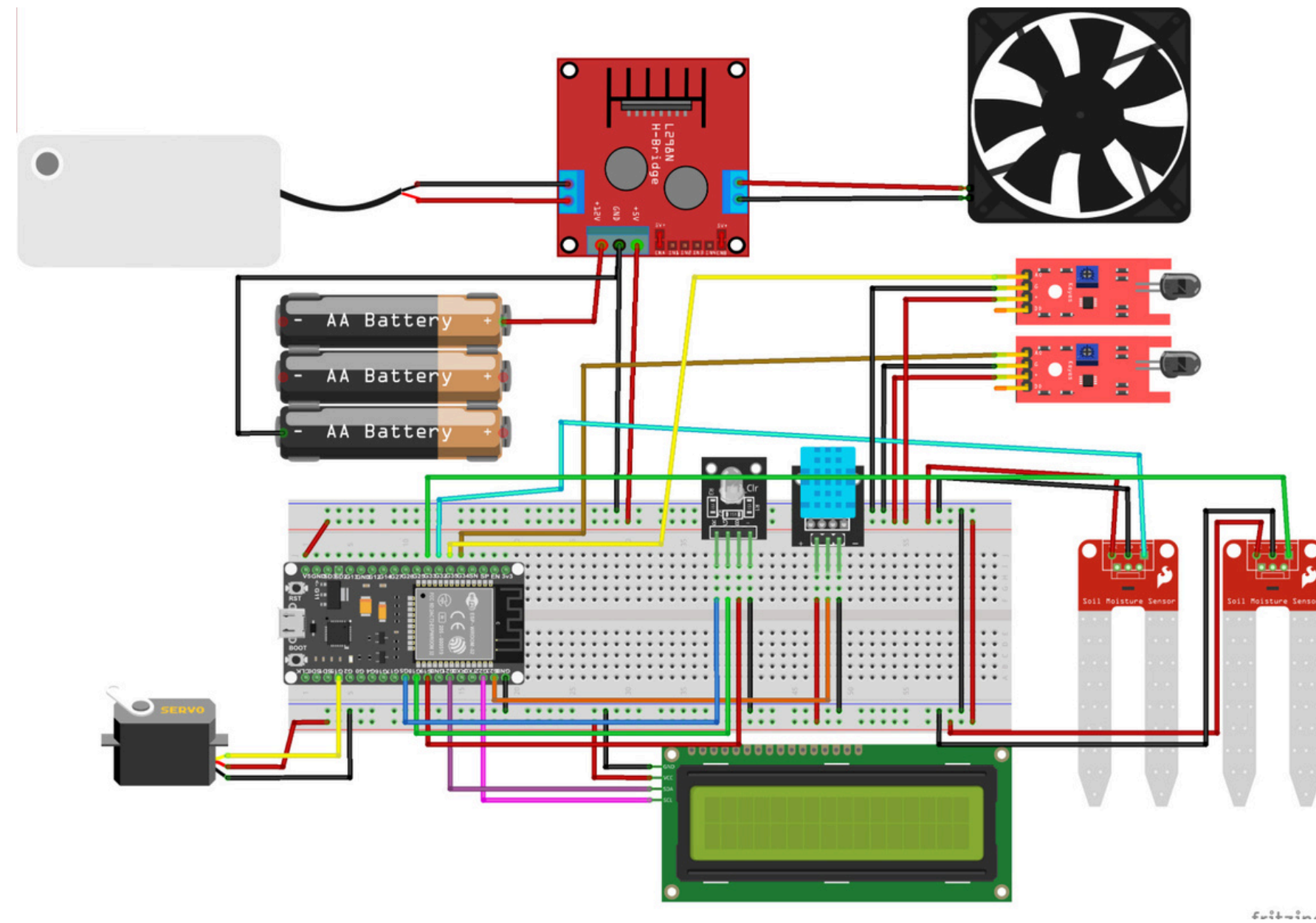


TOOL USED

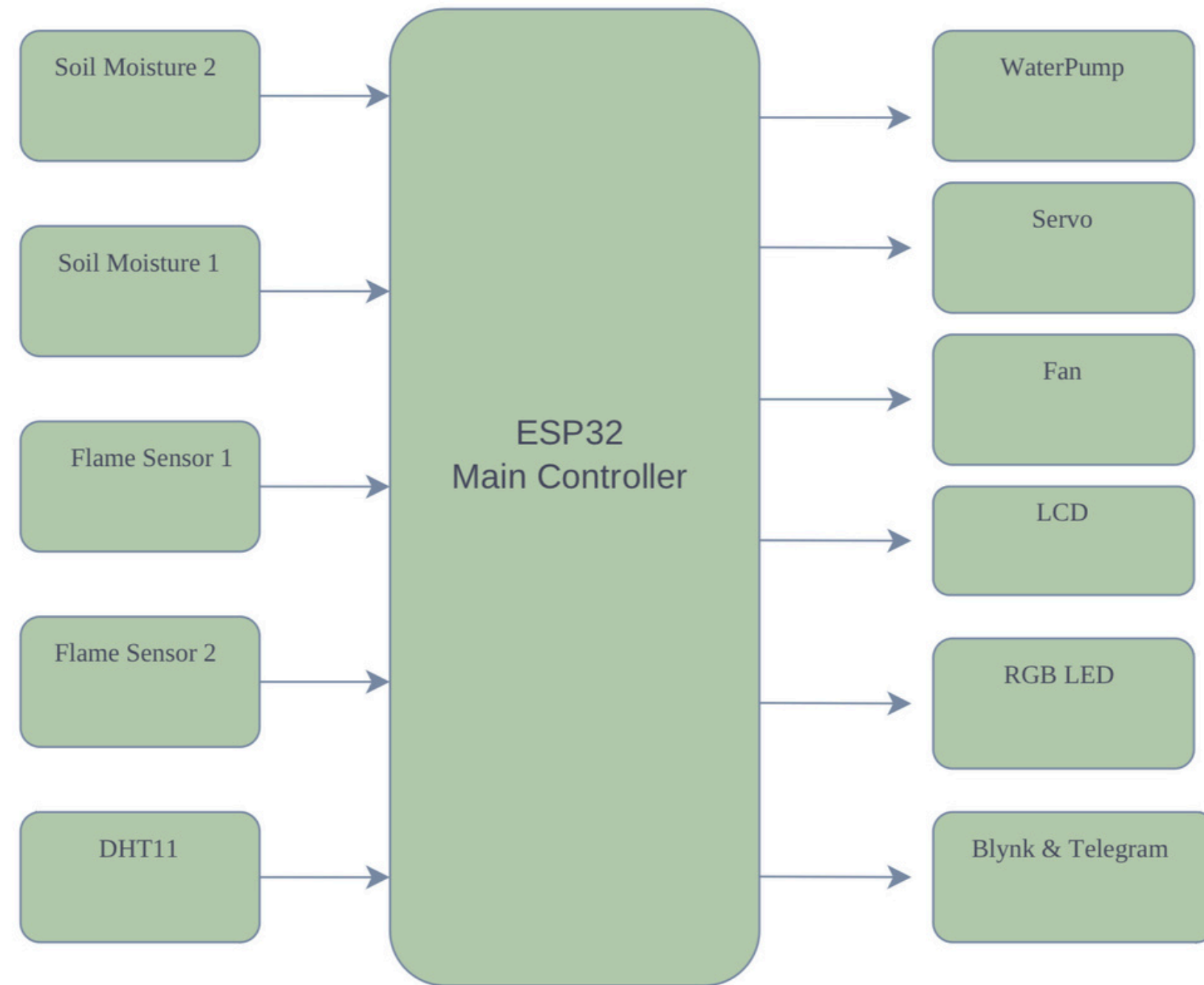
- 18650 charger
- LCD
- RGB LED Module
- Ion rechargeable battery
- L298N Motor Driver Controller Module
- Battery holder



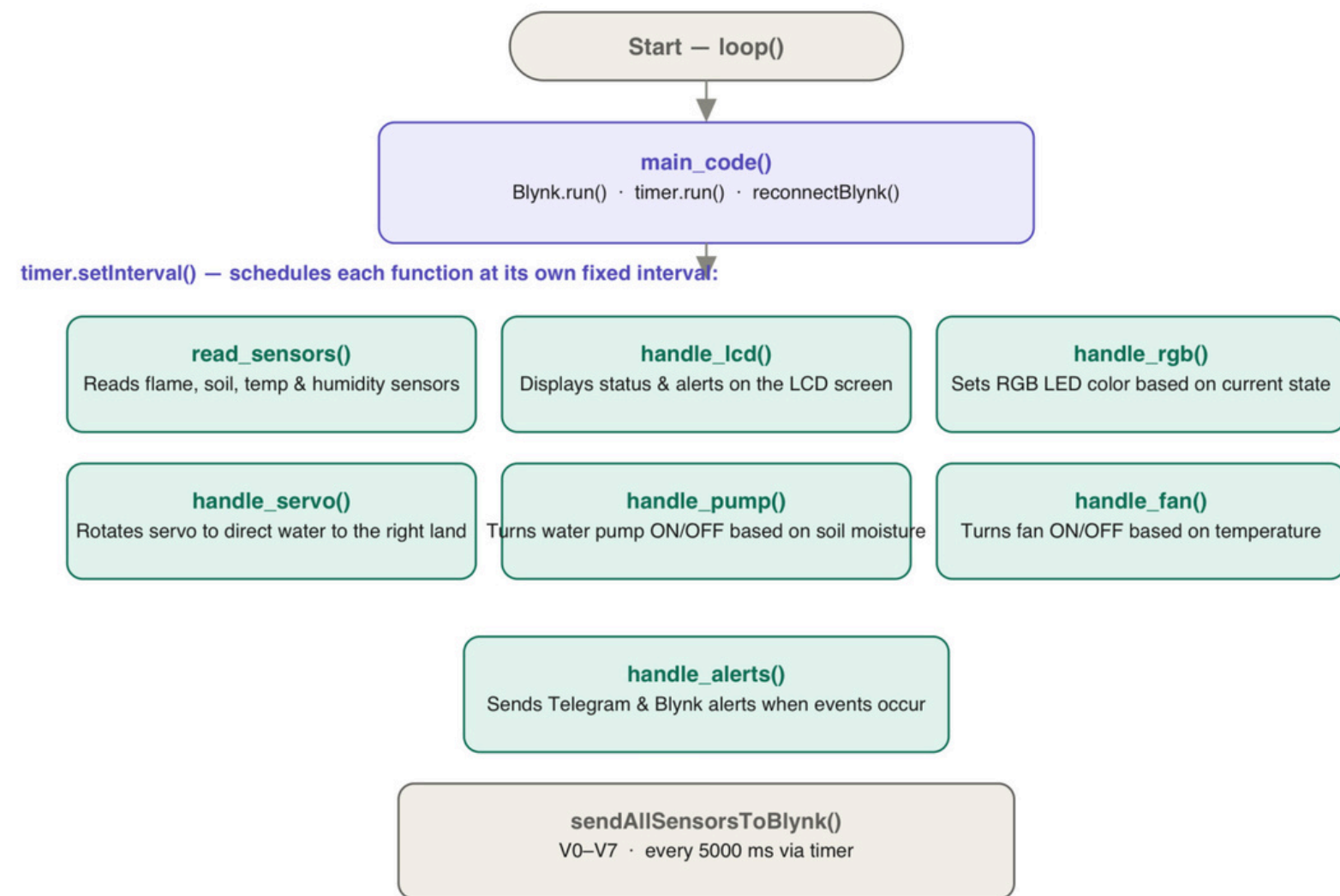
Circuit Diagram



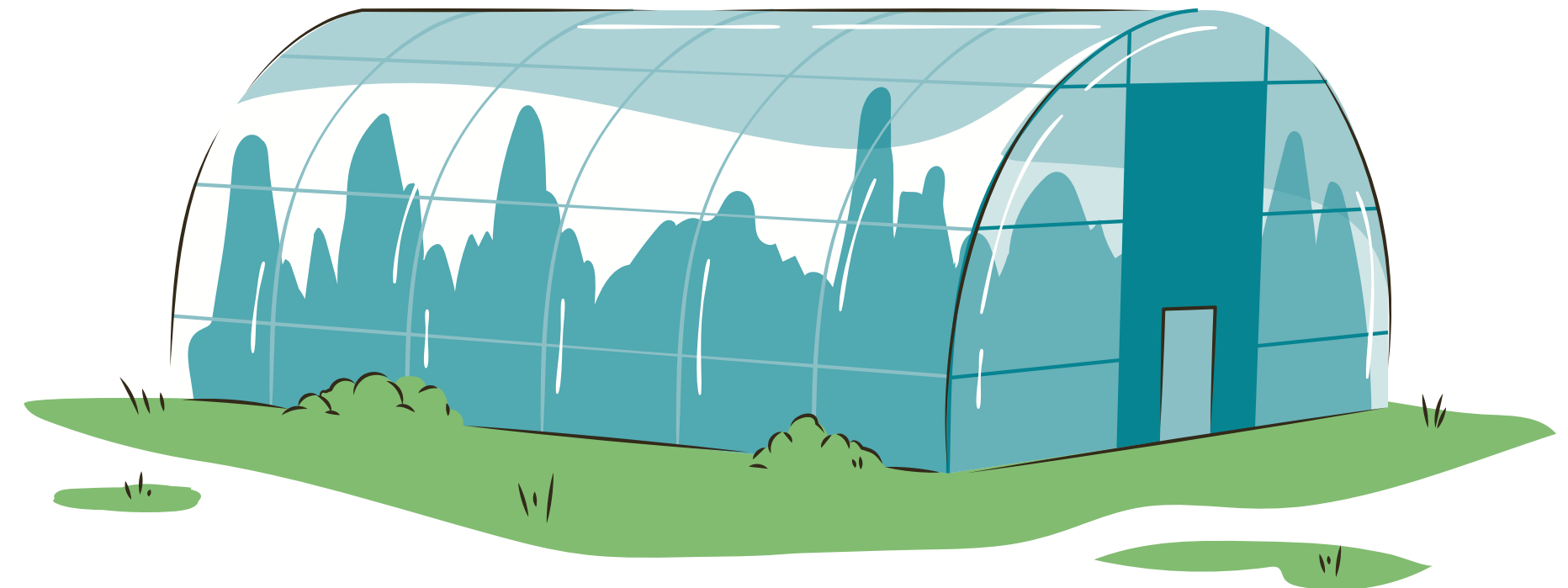
Block Diagram



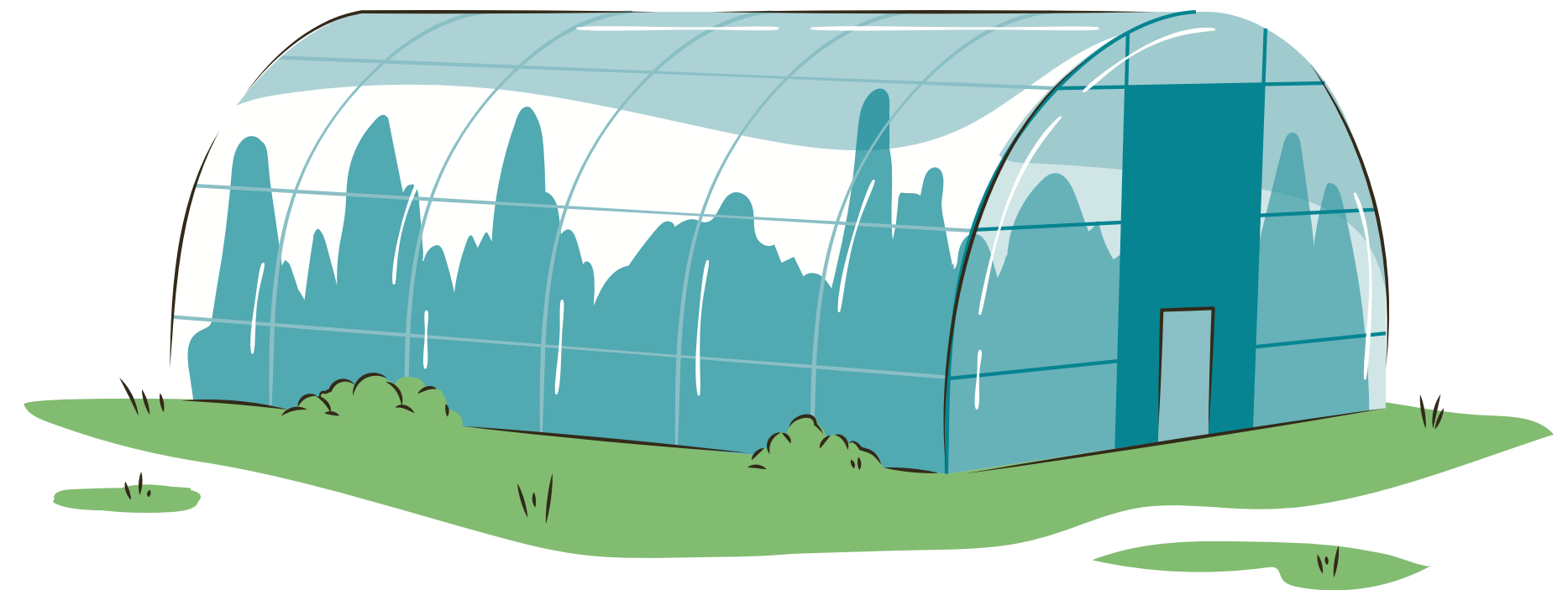
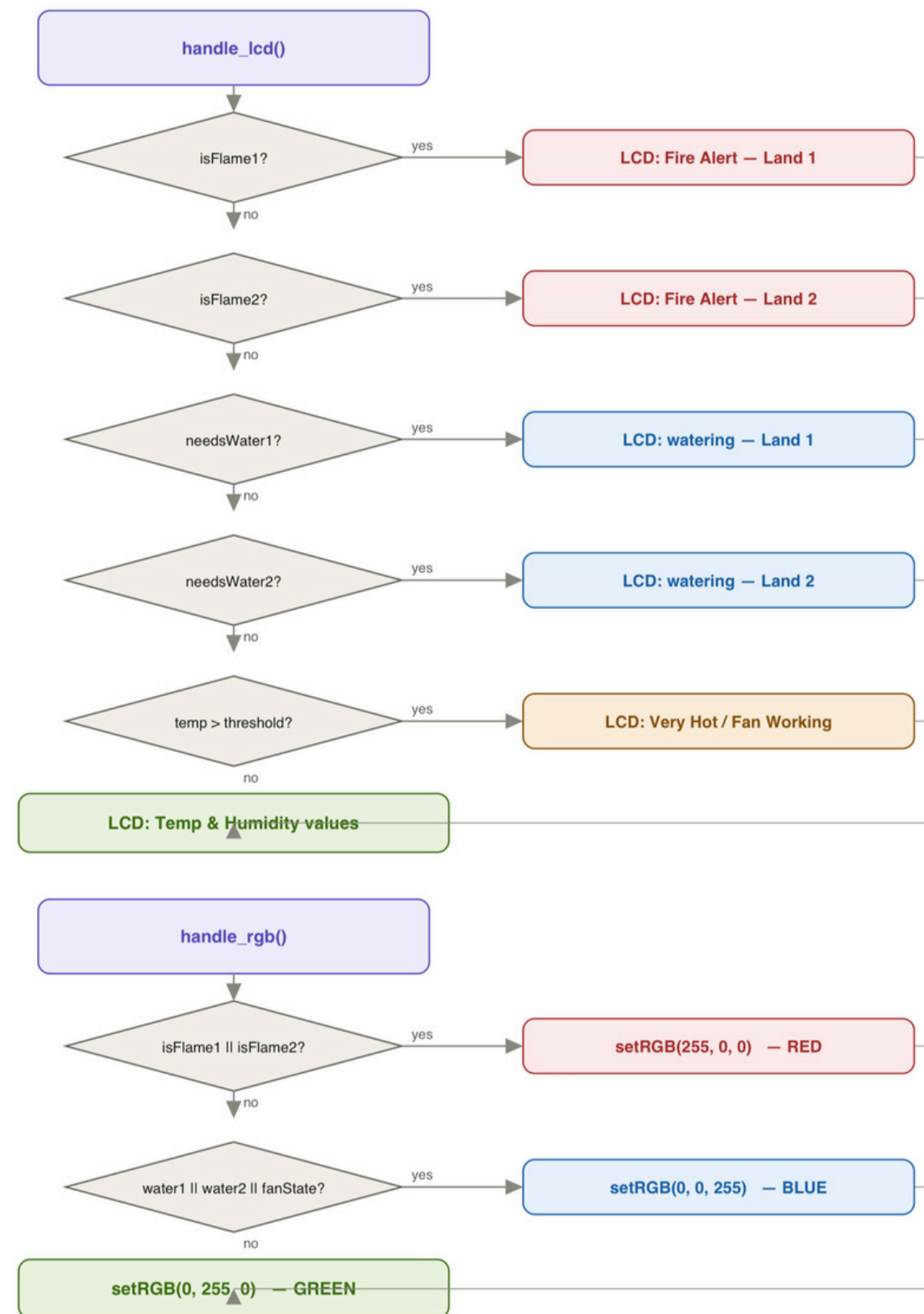
Flowchart



read_sensors() — sub-functions:

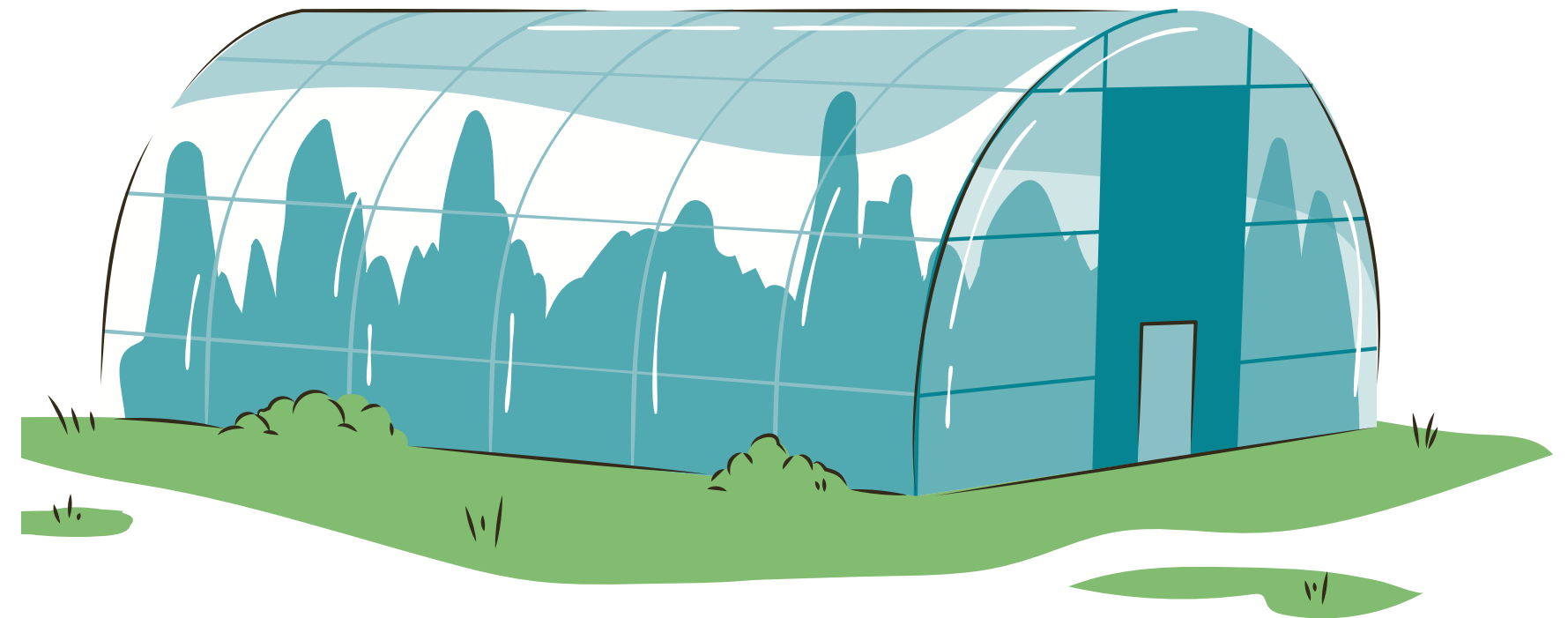
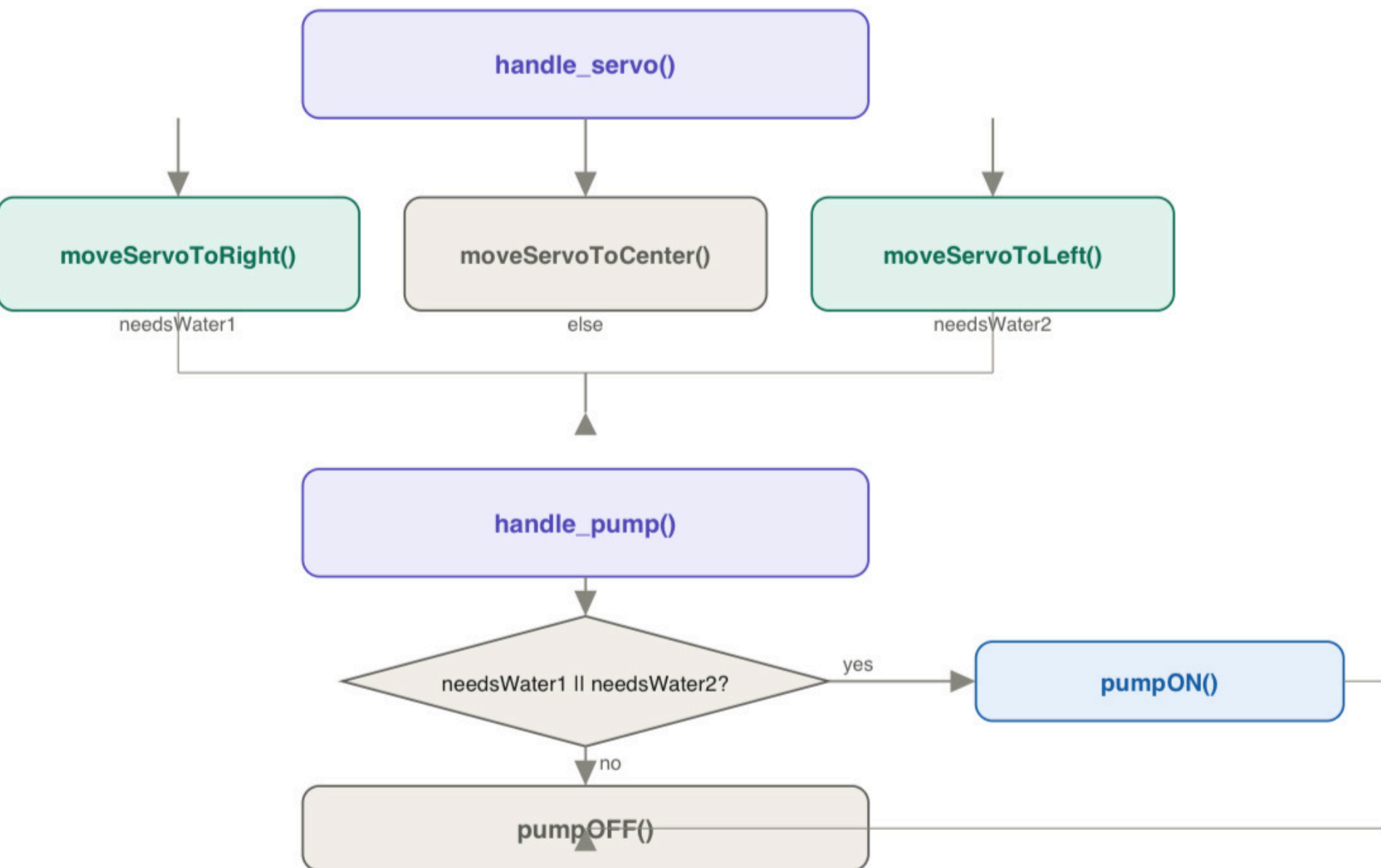


Flowchart

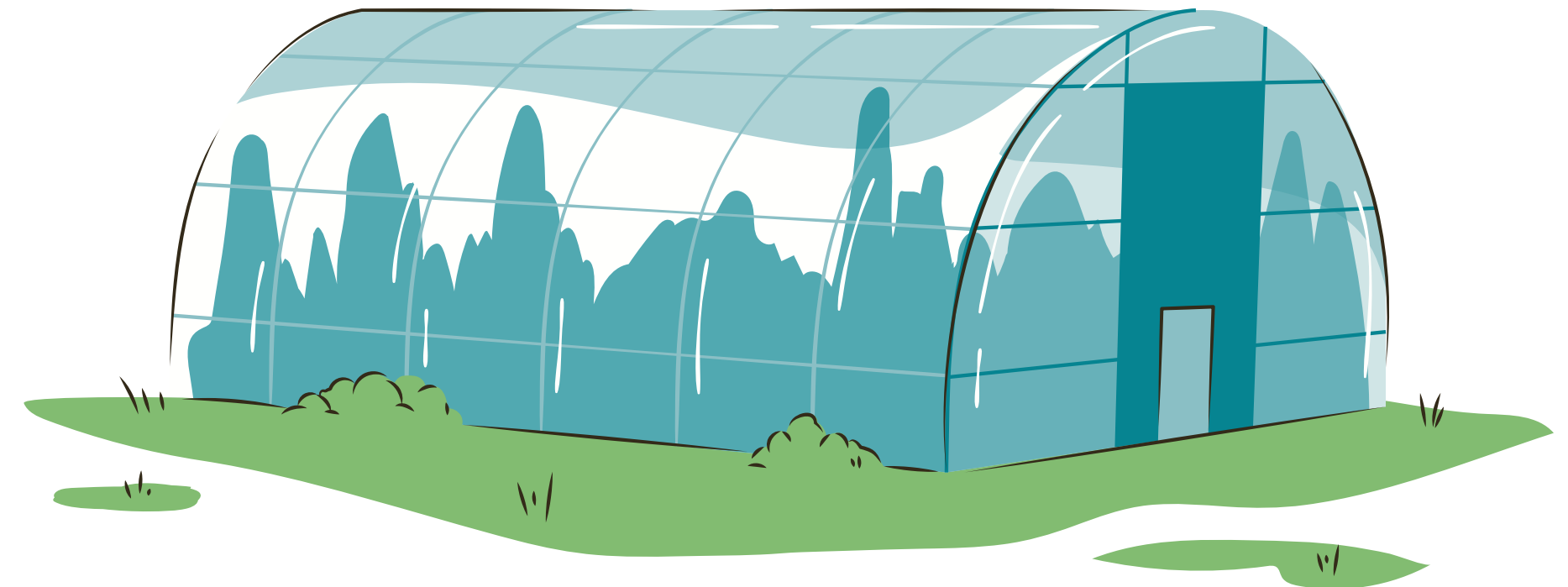
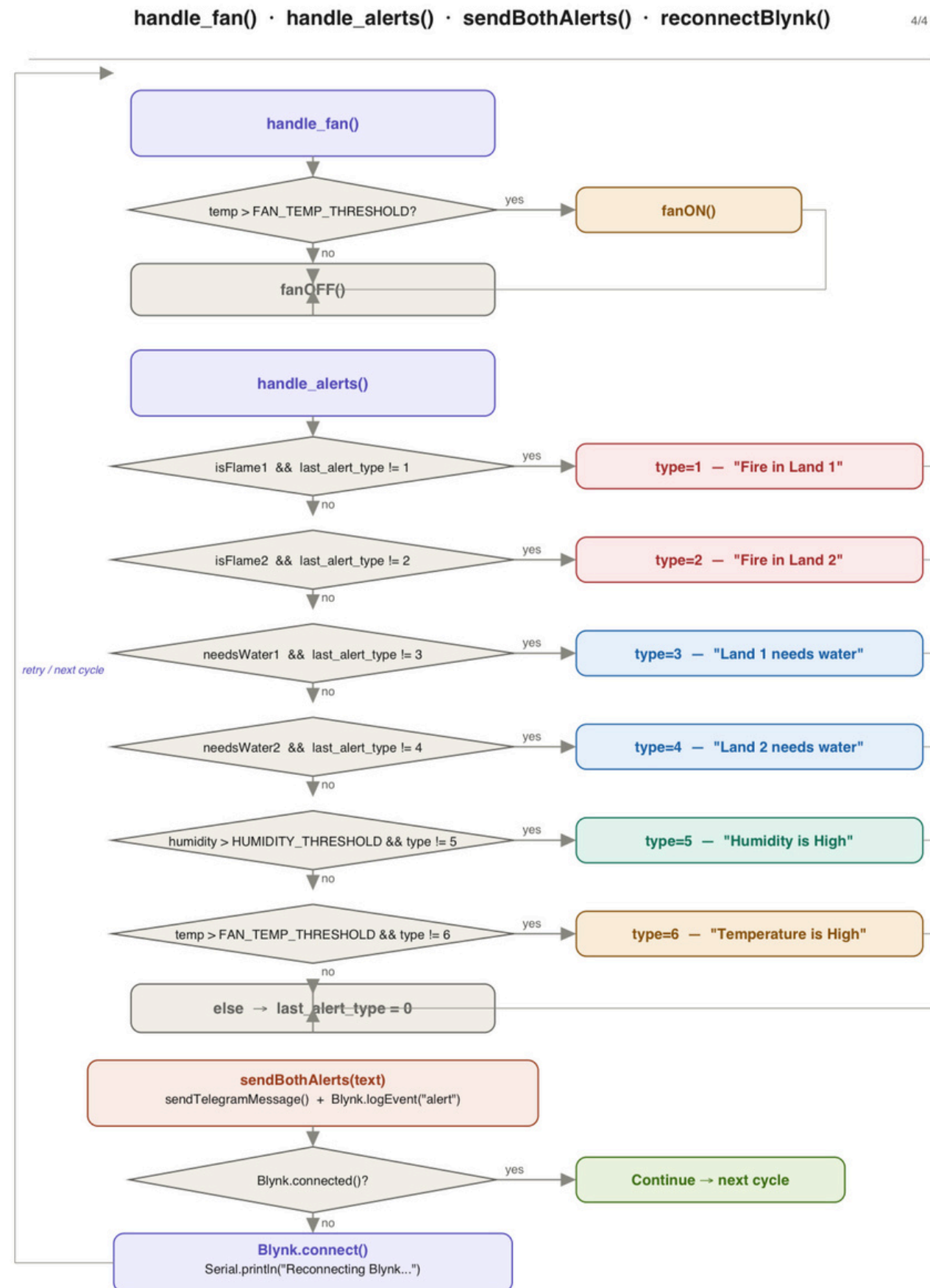


Flowchart

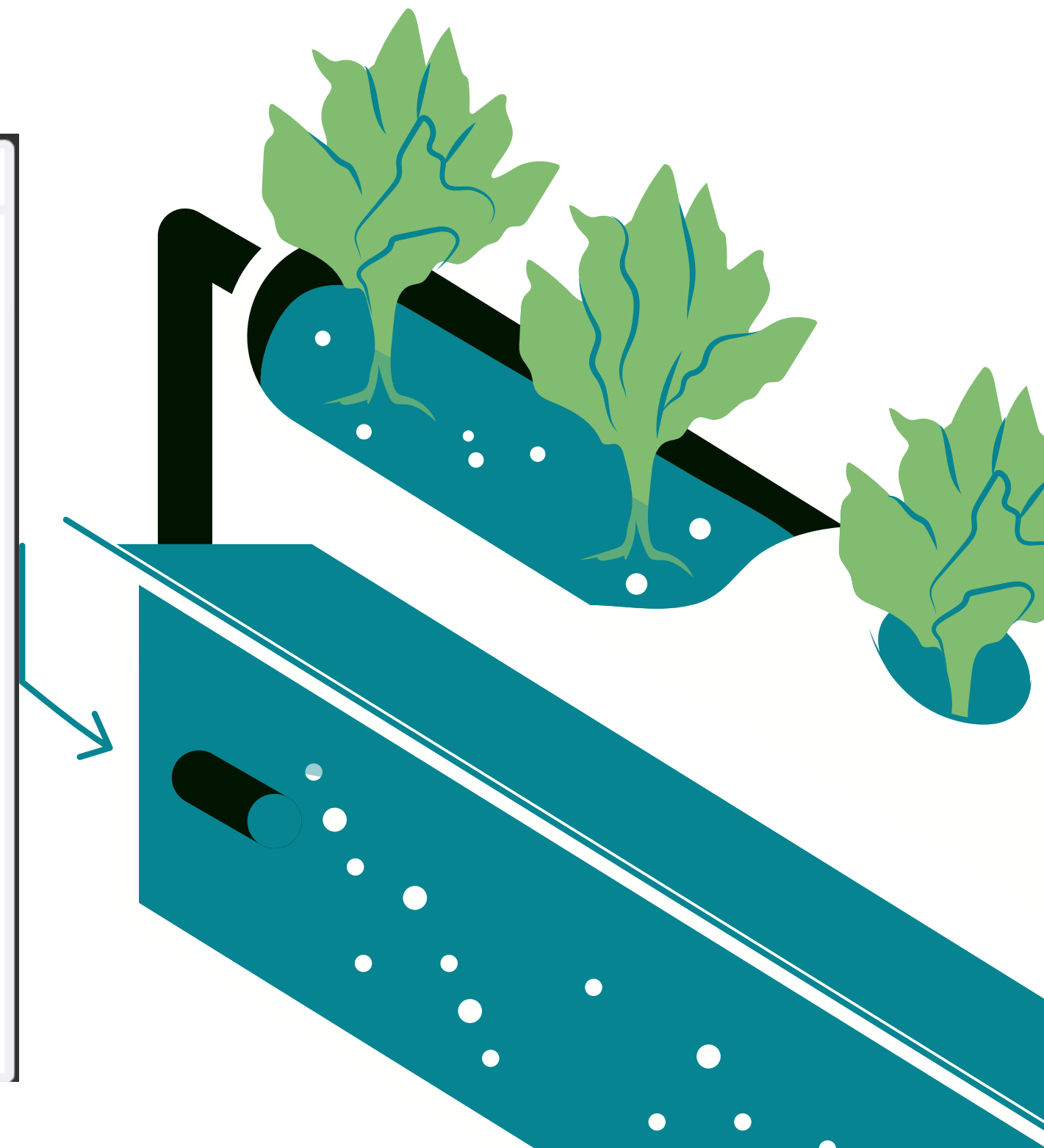
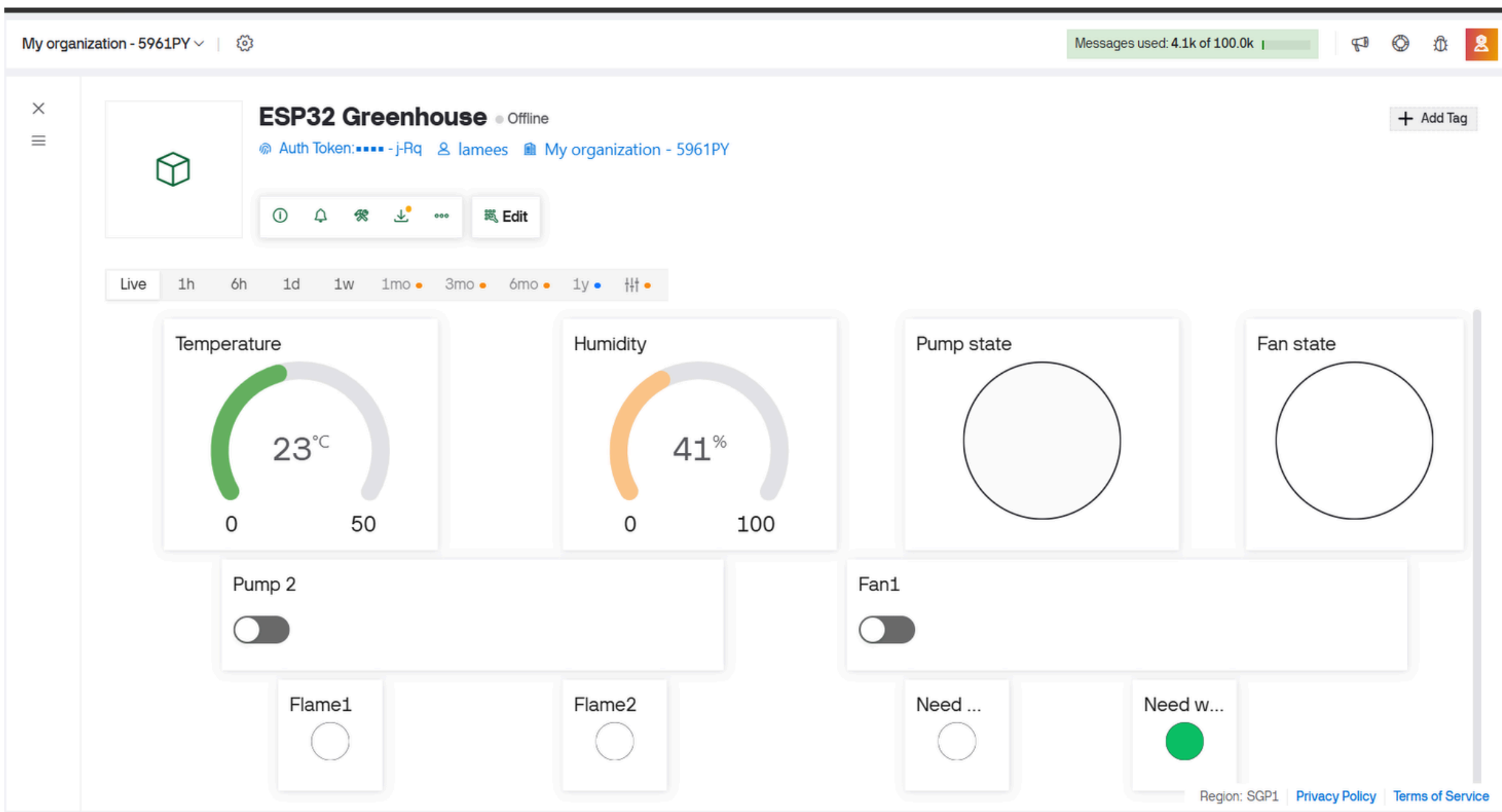
handle_servo() · handle_pump()



Flowchart



Blynk Dashboard



THANK YOU

